

Disentangling land sharing and land sparing effects on bird abundance in France using spatio-temporal Bayesian modeling

Adélie Erard, Ottmar Cronie, Raphaël Lachière-Rey, Romain Lorrillière

February 19, 2026

Abstract

Farmland bird populations have experienced widespread declines across Europe, raising urgent questions about how agricultural landscapes should be structured to reconcile biodiversity conservation and food production. Two contrasting strategies are commonly discussed: land sparing, characterized by intensive agriculture combined with large, separate natural areas, and land sharing, where low-intensity agriculture is embedded within heterogeneous, wildlife-friendly landscapes. However, empirical evidence remains mixed, and few studies have quantified these strategies along a continuous gradient while accounting for spatial and temporal ecological processes.

Using data from the French Breeding Bird Survey (STOC-EPS), we analyzed bird abundance in 810 locations monitored between 2001 and 2024. We quantified a land sharing–sparing gradient using (i) forest area within a 10 km radius, (ii) agricultural land with significant amount of nature, and (iii) a Shannon diversity index of agricultural patches. Bird counts were modeled with negative binomial spatio-temporal Bayesian models fitted using the Integrated Nested Laplace Approximation (INLA), incorporating climate variables, land cover structure, spatial random fields, temporal autocorrelation, and interaction terms. Model selection was performed using WAIC across five species groups: farmland, forest, generalist, and species of conservation concern within farmland and forest habitats.

We found contrasting responses among ecological groups. Forest birds, including species of conservation concern, were positively associated with greater forest area, supporting a land sparing strategy. Generalist species responded positively to higher agricultural heterogeneity, consistent with land sharing benefits. Contrary to expectations, farmland specialists did not benefit from increased agricultural heterogeneity, and farmland species of conservation concern were negatively associated with farmland containing substantial natural elements. Across groups, strong spatial and temporal autocorrelation highlighted the importance of explicitly modeling ecological structure.

Our results demonstrate that no single land-use strategy maximizes abundance across all bird groups. Land sparing appears more favorable for forest specialists and conservation-priority species, whereas land sharing benefits generalists. These findings underscore the need for context-dependent landscape planning and suggest that biodiversity-friendly agricultural practices must be tailored to species ecological requirements rather than relying on a single land management paradigm.